SingleRAN

IP Performance Monitor Feature Parameter Description

Issue Draft A

Date 2025-12-31

HUAWEI TECHNOLOGIES CO., LTD.

Trademarks and Permissions

Notice

Huawei Technologies Co., Ltd.

Address:	Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China
Website:	https://www.huawei.com
Email:	support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
  Changes in functions and their corresponding parameters
- Editorial changes
  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-07-30).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for packet loss detection for IPv6 IP PM sessions. For details, see: • **4.2.3.1 On the Base Station Side** • **4.5.3.1 Data Preparation** • **4.5.3.2 Using MML Commands**	Added the **TRANSFUNCTIONSW**.***PPMIPV6PKTLOSDETSW (5G gNodeB, LTE eNodeB)*** parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Optimized MML command examples. For details, see **4.5.3.2 Using MML Commands**.

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-118607	IP Performance Monitor	4 IP Performance Monitoring
LTE FDD	LBFD-00301201	IP Performance Monitoring	
LTE TDD	TDLBFD-00301201	IP Performance Monitoring	
NB-IoT	MLBFD-12100309	IP Performance Monitoring	
NR	FBFD-010024	IP Performance Monitoring	

3 Overview

Operators using IP transmission are facing the following issues:

- The transmission rate and bandwidth are unstable. Therefore, the operators need to monitor the quality of service (QoS) of transport networks and thereby take appropriate measures to ensure service quality.
- When a transport network becomes faulty, the operators need to identify whether the fault occurs in an NE or on the transport network to isolate the fault.

IP PM uses a Huawei proprietary protocol to monitor the QoS of an end-to-end IP network in real time. Specifically, IP PM monitors the transmission rate, packet loss rate, delay, and delay variation of data streams on the user plane.

Currently, IP PM is supported only over the Abis, Iub, S1, X2, eX2, Xn, eXn, and Se interfaces. The NEs on both ends of monitored data streams must support IP PM.

NOTE

Unless otherwise specified, the S1 and X2 interfaces described in this document include the following:

- S1 interface between an eNodeB and the LTE core network (CN)
- X2 interface between two eNodeBs
- X2 interface between a gNodeB and an eNodeB

It is recommended that IPsec be enabled at the IP layer or ACL rules be added to provide security protection for IP PM.

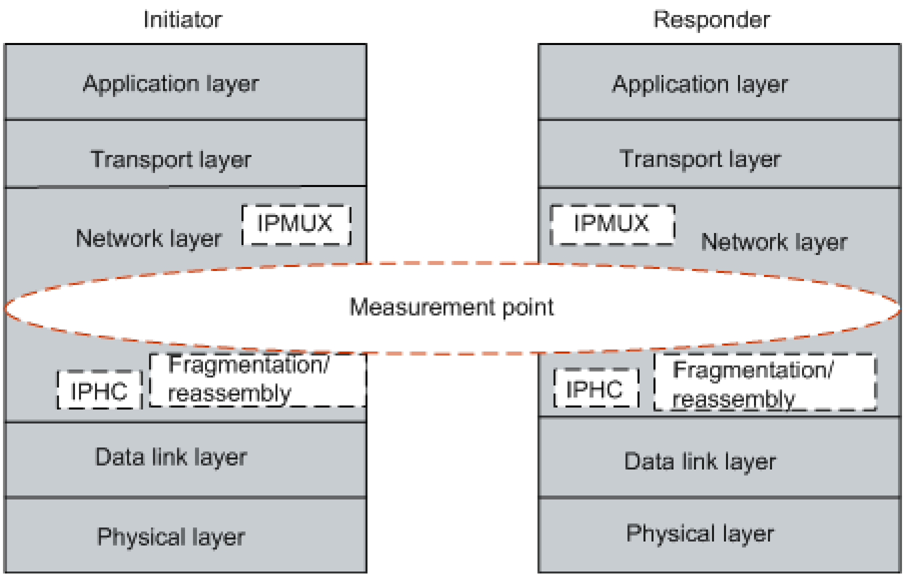
4 IP Performance Monitoring

4.1 Principles

4.1.1 Location of IP PM in the TCP/IP Protocol Model

Figure 4-1 shows the location of IP PM in the TCP/IP protocol model.

Figure 4-1 Location of IP PM in the TCP/IP protocol model



As shown in **Figure 4-1**, IP PM is implemented at the network layer. Specifically, it is implemented below the IP MUX sublayer and above the IP Header Compression (IPHC) and IP packet fragmentation/reassembly.

The IP PM initiator measures the packet loss rate, delay, and delay variation of a transport network after IP multiplexing. Then, the initiator compresses and fragments the IP packets. The IP PM responder measures the packet loss rate, delay, and delay variation of a transport network after decompressing and reassembling the IP packets. Then, the responder performs IP demultiplexing.

4.1.2 Concepts

4.1.2.1 Detected Data Stream

IP PM calculates packets included in a data stream and measures the transmission rate, packet loss rate, and other relevant information of the data stream. IP PM uses triplets or four-tuples to identify a data stream. The triplets are source IP (SIP), destination IP (DIP), and protocol type (PT). The four-tuples are SIP, DIP, PT, and differentiated services code point (DSCP).

- SIP: source IP addresses of service packets to be measured and of IP PM frames
- DIP: destination IP addresses of service packets to be measured and of IP PM frames
- PT: protocol type of packets on data streams, indicating the type of packets to be measured. The type of Huawei IP PM packets is UDP and is not configurable.
- DSCP: differentiated services code point values carried in service packets to be measured and in IP PM frames

  If a detected data stream is identified by four-tuples, IP PM calculates the delay and packet loss rate of IP PM packets based on the DSCP values carried in IP packet headers. Therefore, the intermediate transmission equipment must not change the DSCP values. DSCP values on transmission links are detected by an NE when IP PM is activated. Once detecting that the DSCP value on the transmit end is different from that on the receive end, the NE reports an IP PM activation failure alarm with the cause of DSCP value being changed.

Whether to identify a detected data stream by triplets or four-tuples depends on the priority of the data stream. Triplets apply if the data stream priority can be neglected, whereas four-tuples apply if the data stream priority must be considered.

Currently, four-tuples are supported by all NEs and triplets are supported only by the eGBTS, NodeB, eNodeB, gNodeB, co-MPT multimode base station, and BSC6960. The ***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** and **IPPOOLPM**.***PHB*** parameters determine whether triplets or four-tuples are used for the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station and BSC6960, respectively. The latter parameter specifies the priority of services with different DSCPs. The value **ALL** indicates any service priority, which indicates triplets. If triplets are used, both ends must support this IP PM type; otherwise, IP PM will fail to be activated.

The data stream to be measured can be specified in multiple ways based on the NE type, as listed in the following table.

NE	Directly Specifying a Data Stream	Bound to an IP Path	Bound to a Transmission Resource Pool
BSC6900	Not supported	Supported	Supported

NE	Directly Specifying a Data Stream	Bound to an IP Path	Bound to a Transmission Resource Pool
BSC6910/BSC6960	Not supported	Not supported	Supported
Base stations	Supported	Supported	Not supported

- Directly specifying a data stream

  SIP, DIP, and DSCP are configured to directly specify a detected data stream.
- Binding to an IP path

  IP PM is bound to an IP path so that local and peer IP addresses of the IP path are used as the SIP and DIP, respectively. In this way, IP PM detects data streams on this IP path.

  When a detected data stream is identified by four-tuples, the DSCP of the detected data stream is either of the following:
  – DSCP configured for the IP path
  – DSCP configured for IP PM, if no DSCP is configured for the IP path
- Binding to a transmission resource pool

  IP PM is bound to a transmission resource pool so that the local address in the resource pool is used as the SIP and the peer address (address of the ANI) is used as the DIP.

4.1.2.2 IP PM Frame

IP PM defines ACT, ACT-ACK, forward monitoring (FM), backward reporting (BR), DEA, DEA-ACK frames.

- ACT and ACT-ACK frames are used to activate an IP PM session.

  Two ends of IP PM interact with each other through the ACT and ACT-ACK frames for establishing IP PM links.

  An ACT frame is sent by the IP PM initiator to the IP PM responder to activate an IP PM session.

  An ACT-ACK frame is sent by the IP PM responder to respond to an ACT frame from the IP PM initiator for acknowledging activation.
- FM and BR frames are used for IP PM detection.

  Two ends of IP PM exchange information about their sent and received packets with each other through the two types of frames for detecting the QoS of an IP link.

  Both FM and BR frames can be sent by either the initiator or the responder.

  An FM frame is a forward frame for monitoring QoS performance. It is periodically sent by an IP PM end.

  A BR frame is backward frame for monitoring QoS performance. It is periodically sent by the IP PM end upon receiving an FM frame.
- DEA and DEA-ACK frames are used to deactivate an IP PM session.

Two ends of IP PM interact with each other through the DEA and DEA-ACK frames for disconnecting IP PM links.

A DEA frame is sent by the initiator to the responder, notifying the responder that an IP PM session is to be deactivated.

A DEA-ACK frame, used for acknowledging deactivation, is sent by the responder to respond to a DEA frame from the initiator.

4.1.2.3 IP PM Activation Direction

IP PM defines the following activation directions:

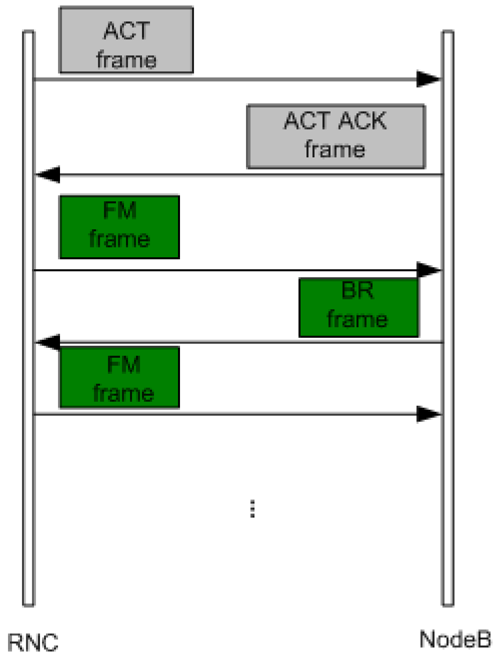
- Forward activation

  The forward activation measures the performance of transmission from the initiator to the responder.

  The ACT and FM frames are sent in the same direction, that is, both are sent from the initiator to the responder.

  For example, in the case of forward IP PM activation by an RNC over the Iub interface, both ACT and FM frames are sent from the RNC to a NodeB and then the NodeB responds to the RNC with ACT-ACK and BR frames.

  Figure 4-2 Forward IP PM activation by an RNC over Iub



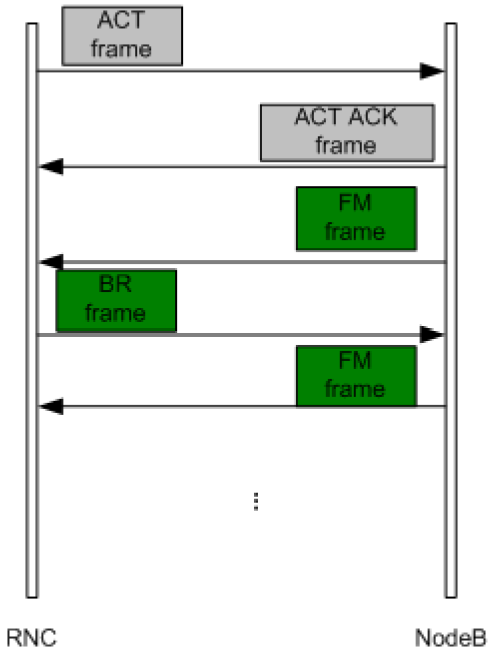
- Backward activation

  The backward activation measures the performance of transmission from the responder to the initiator.

  The ACT and FM frames are sent in opposite directions, that is, the initiator sends an ACT frame and the responder sends an FM frame.

  For example, in the case of backward IP PM activation by an RNC over the Iub interface, the RNC sends an ACT frame to a NodeB, but the NodeB sends an FM frame to the RNC and the RNC responds to the FR frame with a BR frame.

Figure 4-3 Backward IP PM activation by an RNC over Iub

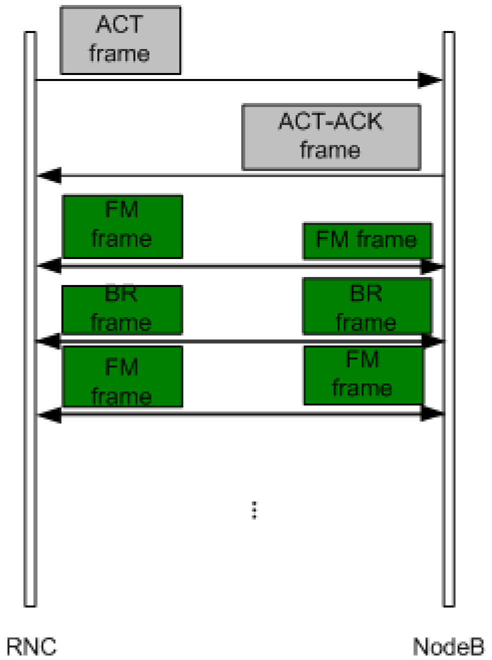


- Bidirectional activation

  Both the initiator and the responder can send an FM frame or a BR frame.

  For example, in the case of bidirectional IP PM activation by an RNC over the Iub interface, the RNC sends an ACT frame to a NodeB, and both the NodeB and RNC can send an FM frame to each other.

Figure 4-4 Bidirectional IP PM activation by an RNC over Iub



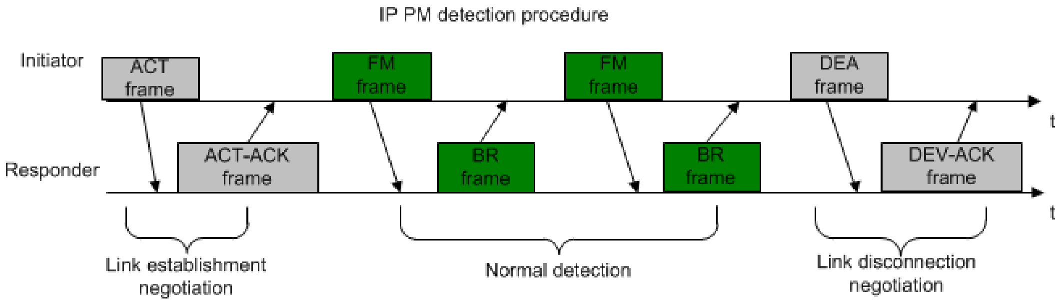
Backward and bidirectional activations are only recommended when transmission resource pool networking is applied for base station controllers.

4.1.3 IP PM Detection

IP PM detection is split into three stages: link establishment negotiation, normal detection, and link disconnection negotiation.

Figure 4-5 uses forward activation as an example for IP PM detection procedure.

Figure 4-5 IP PM detection procedure (forward activation)



The detailed procedure for IP PM detection (forward activation) is as follows:

1. Link establishment negotiation: to establish an IP PM link

   The IP PM initiator sends an ACT frame to the IP PM responder. The ACT frame carries information about a detected data stream, including the source IP address, destination IP address, activation direction, and other information. The IP PM responder responds to the ACT frame with an ACT-ACK frame to complete the negotiation on the detected data stream. The ACT-ACK frame carries information indicating activation success or failure.

2. Normal detection: to perform IP PM detection after the IP PM link is successfully established

   The initiator periodically sends the responder an FM frame. This FM frame records T1 (the time when the FM frame is sent) and the number of real-time service packets and bytes sent within the current measurement period. Upon receiving an FM frame, the responder responds with a BR frame. The BR frame records the number of real-time service packets and bytes transmitted and received within the current measurement period, T1 (the time the FM frame is sent), T2 (the time when the FM frame is received), and T3 (the time when the BR frame is sent).

   The initiator records T4 (the time when the BR frame is received) and calculates the measurement parameters related to IP PM within the current measurement period, such as the delay, transmission rate, packet loss rate, and delay variation. For details about when to use and how to calculate these measurement parameters, see **4.1.4 Related Measurement Parameters**.

3. Link disconnection negotiation: to disconnect an IP PM link after the detection is completed

   – The initiator sends a DEA frame.

   – Upon receiving a DEA frame, the responder responds to the initiator with a DEA-ACK frame.

4.1.4 Related Measurement Parameters

4.1.4.1 Delay

IP PM measures the round trip delay, which refers to the sum of the delays in transmitting an FM frame and a BR frame in a round trip between the initiator and the responder.

Round trip delay can be calculated using the following formula in IP PM of forward or bidirectional activation:

Round trip delay = (T2 – T1) + (T4 – T3)

where

- T1: Time when an FM frame is sent
- T2: Time when the FM frame is received
- T3: Time when a BR frame is sent
- T4: Time when the BR frame is received (recorded by the initiator)

Note that T1 and T2 are recorded in the BR frame received by the initiator.

IP PM supports one-way delay measurement when coordinated services are performed over the eX2/uX2/eXn interface and the two NEs are time synchronized. The one-way delay equals T2 minus T1.

4.1.4.2 Transmission Rate

IP PM calculates the transmission rate between two ends. The base station checks the transmission rate every measurement period (1s), and the base station controller checks the transmission rate every measurement period (5s).

IP PM obtains the numbers of real-time service packets and bytes that are sent and received in the current measurement period, based on the information carried in FM and BR frames. Then, IP PM calculates the following items using these numbers:

- Rate in transmitting packets = Number of sent packets in the measurement period/Measurement period
- Rate in receiving packets = Number of received packets in the measurement period/Measurement period
- Rate in transmitting bits = Number of sent bytes in the measurement period x 8/Measurement period
- Rate in receiving bits = Number of received bytes in the measurement period x 8/Measurement period

4.1.4.3 Packet Loss Rate

When IP PM is enabled, the packet loss rate between two ends is calculated based on the numbers of sent and received packets recorded in FM and BR frames.

In forward activation, the IP PM initiator calculates the packet loss rate on the path from the initiator to the responder based on the numbers of real-time service packets that are sent and received in the current measurement period, which are recorded in the BR frame received by the initiator. The formula for calculating the packet loss rate is as follows:

Packet loss rate = (Number of sent packets – Number of received packets)/ Number of sent packets

In backward activation, the IP PM initiator calculates the packet loss rate on the path from the responder to the initiator based on numbers of real-time service packets that are sent and received in the current measurement period, which are recorded in the FM frame received by the initiator.

Parameters can be configured to control whether the NE reports an alarm when the packet loss rate exceeds a specified threshold. For details, see **4.2 Applications**.

NOTE

When there are no ongoing services, the packet loss rate is invalid or zero, which is meaningless of reference.

4.1.4.4 Delay Variation

IP PM calculates the delay variation between two ends based on the delay in transmitting FM and BR frames.

The forward delay and backward delay can be calculated based on the following:

- T1: Time when an FM frame is sent
- T2: Time when the FM frame is received
- T3: Time when the BR frame is sent
- T4: Time when the BR frame is received (recorded by the initiator)

The forward and backward delay variations are calculated based on differences in delays of two consecutive FM and BR frames.

4.1.5 IP PM Version

IP PM has four versions: V1, V2, V14, and V15. The boards that support IP PM V1 are listed in **Table 4-10**.

IP PM V2 enhances IP PM V1 and will replace IP PM V1 on live networks. The enhancements are as follows:

- Modified calculation method: Corrects inaccuracy in IP PM statistics in the case of out-of-order service packets.
- Modified DSCP detection scheme: An NE performs DSCP detection on measured transmission links during IP PM detection. Upon detecting that the DSCP value on the transmit end is different from that on the receive end, the NE reports an IP PM activation failure alarm, with the cause value of DSCP value being changed. In this case, the IP PM session will be deactivated.
- Optimized alarm description: Provides the activation failure cause in the related alarm.

Currently, the GOUa/FG2a/GTMUc does not support IP PM V2.

IP PM V15 is supported from SRAN10.1 onwards. IP PM V15 supports delay measurement with microsecond precision and session self-establishment over the eX2/eXn interface. Currently, only the UMPT/UTRPc/UBBPe/UBBPg supports IP PM V15.

IP PM V14 is supported from SRAN12.0 onwards. IP PM V14 supports one-way delay measurement. IP PM V14 sessions can be set up when coordinated services are performed over the eX2/uX2/eXn interface. Currently, only the UMPT/UTRPc supports IP PM V14.

IP PM versions may be inconsistent on NEs of two ends on live networks. If a version inconsistency exists, IP PM packets are sent and calculated based on an earlier IP PM version.

4.2 Applications

4.2.1 Overview

IP PM can be applied to the Abis/Iub interface between a base station and a base station controller, X2/eX2 interface between eNodeBs, S1 interface between an eNodeB and the EPC, X2 interface between a gNodeB and an eNodeB in non-standalone (NSA) NR architecture, and Xn/eXn interface between two gNodeBs in SA NR architecture.

For details about the application of IP PM, see **4.1.3 IP PM Detection**.

NOTE

The application of IP PM on a co-MPT multimode base station is not described in this document because it is the same as that on an eGBTS, NodeB, eNodeB, or gNodeB.

IP PM and IP Active Performance Measurement each has its own inherent strengths and limitations and they two can help balance each other with complementary strengths. For details about their application differences, see *IP Active Performance Measurement*.

4.2.2 IP PM Between the Base Station Controller and Base Station

This section describes the application of IP PM between the base station controller and base station.

4.2.2.1 On the Base Station Controller Side

When connecting to a base station, a base station controller (BSC6900, BSC6960, or BSC6910) acts as the initiator to activate the IP PM. IP PM configuration and performance measurement vary in transmission resource pool and non-transmission resource pool networking.

For details on the transmission resource pool, see *Transmission Resource Pool in BSC* in *GBSS Feature Documentation* and *Transmission Resource Pool in RNC* in *RAN Feature Documentation*.

Table 4-1 provides whether a base station controller supports IP PM in different networking scenarios.

Table 4-1 Whether a base station controller supports IP PM

Scenario	GSM	UMTS
BSC6900 in non-transmission resource pool networking	Supported	Supported
BSC6900 in transmission resource pool networking	Not supported	Supported
BSC6910/BSC6960 in transmission resource pool networking	Supported	Supported

NOTE

The BSC6910/BSC6960 only supports transmission resource pool networking.

4.2.2.1.1 IP PM in Non-Transmission Resource Pool Networking

The BSC6900 supports non-transmission resource pool networking.

You can run the BSC6900 MML command **ACT IPPM** to activate IP PM. **Table 4-2** describes the items you need to configure in this command.

NOTE

- Before activating IP PM on the base station controller, ensure that **TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW*** has been set to **ENABLE** on the base station. If this parameter is not set to **ENABLE**, the base station cannot respond to IP PM sessions.
- If the base station is a GBTS, ensure that **BTSGTRANSPARA**.***IPPMADMITTANCE*** has been set to **ENABLE** before activating IP PM on the base station controller. If this parameter is not set to **ENABLE**, the GBTS cannot respond to IP PM sessions.

Table 4-2 Items and configuration principles for activating IP PM on the BSC6900 in a non-transmission resource pool networking

Item	Description
Detected data stream	For details on the definition of a detected data stream, see **4.1.2.1 Detected Data Stream**. A detected data stream can only be specified by the ***PATHID*** parameter, which specifies the IP path to which IP PM is to be bound. ● For an IP path of QoS type, the DSCP of the detected data stream is the DSCP configured for IP PM. ● For an IP path of non-QoS type, the DSCP of the detected data stream is the DSCP configured for the IP path. You can run the **LST IPPATH** command to check whether an IP path is a QoS path or not. The ***ISQOSPATH*** parameter specifies whether an IP path is a QoS path.
Activation direction	Only forward activation can be used and the activation direction is not configurable.
FM packet send period	The sending packet period, or the period for sending FM frames is specified by the ***PMPRD*** parameter.
Reporting an alarm once the predefined packet loss rate threshold is exceeded	The ***LOSTPKTDETECTSW*** parameter specifies whether to enable the function of reporting an alarm once the predefined packet loss rate threshold is exceeded. When the packet loss rate exceeds the threshold specified by the ***LOSTPKTALARMTHD*** parameter, ALM-21352 IP Path Excessive Packet Loss Rate is reported.

4.2.2.1.2 IP PM in Transmission Resource Pool Networking

The BSC6900, BSC6960, and BSC6910 support the transmission resource pool networking. For the BSC6900 in the transmission resource pool networking, IP PM can be activated only over the Iub interface.

You can run the **ACT IPPOOLPM** command on the BSC6900, BSC6960, or BSC6910 to activate IP PM. **Table 4-3** describes the items you need to configure through this command.

NOTE

- Before activating IP PM on the base station controller, ensure that **TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW*** has been set to **ENABLE** on the base station. If this parameter is not set to **ENABLE**, the base station cannot respond to IP PM sessions.
- If the base station is a GBTS, ensure that **BTSGTRANSPARA**.***IPPMADMITTANCE*** has been set to **ENABLE** before activating IP PM on the base station controller.

Table 4-3 Items and configuration principles for activating IP PM on the BSC6900/BSC6910/BSC6960 in transmission resource pool networking

Item	Description
Detected data stream	For details on the definition of a detected data stream, see **4.1.2.1 Detected Data Stream**. For the BSC6900/BSC6910 in transmission resource pool networking, a detected data stream can only be specified by being bound to a transmission resource pool. The ***SIPTYPE*** parameter is used to specify which local IP address in the transmission resource pool is used as an SIP as follows: • If ***SIPTYPE*** is set to **NODEB_BIND_SIP(Nodeb Bind Sip)**, all IP addresses bound to the adjacent node are used as SIPs, that is, detected data streams are the ones from all IP addresses bound to the adjacent node to their peer addresses. • If ***SIPTYPE*** is set to **ASSIGN_ONE_SIP(Assign One Sip)**, a specific IP address is used as the SIP, that is, the detected data stream is the one from the specified IP address to its peer address. • For the Abis interface, ***SIPTYPE*** can only be set to **ADJNODE_BIND_SIP**. The DSCP of the data stream to be detected can be obtained from the ***PHB*** parameter. For details about mapping between the **PHB** parameter and DSCP, see *Transmission Resource Management*. The BSC6960 measures the data streams from an IP address in the IP transmission resource pool bound to an adjacent node to the peer IP address.
Activation direction	The activation direction is specified by the ***DR*** parameter, whose values **SOURCE**, **SINK**, and **BOTH** indicate forward activation, backward activation, and bidirectional activation, respectively. • For the Abis interface, only forward activation is supported and the activation direction is not configurable. • For the Iub interface, forward, backward, and bidirectional activation can all be configured.
FM packet send period	The sending packet period, or the period for sending FM frames is specified by the ***PMPRD*** parameter.
Reporting an alarm once the predefined packet loss rate threshold is exceeded	The ***LOSTPKTDETECTSW*** parameter specifies whether the function of reporting an alarm once the predefined packet loss rate threshold is exceeded is enabled. When this function is enabled and the packet loss rate exceeds the parameter value, the BSC6900/BSC6910 reports ALM-21395 Adjacent Node IP Path Excessive Packet Loss Rate and the BSC6960 reports ALM-21695 Adjacent Node IP Path Excessive Packet Loss Rate.

NOTE

For a base station controller in transmission resource pool networking, if its interface board does not support the number of IP PM sessions planned for base stations, the excess IP PM sessions cannot be activated. For details about the number of IP PM sessions supported by base station controller boards, see Table 4-10. Under these circumstances, run the **ACT IPPOOLPM** command on the base station controller to activate IP PM sessions.

- If the performance of the IP connection between the SIP of the base station controller and the base station is monitored, set the ***SIPTYPE*** parameter to **NODEB_BIND_SIP** for the BSC6900 and to **ADJNODE_BIND_SIP** for the BSC6910.
- If the performance of the IP connection between a specific SIP in the transmission resource pool and the base station is monitored, set the ***SIPTYPE*** parameter to **ASSIGN_ONE_SIP**.
- It is recommended that IP PM sessions using triplets be activated on the BSC6960.

4.2.2.2 On the Base Station Side

When connecting to a base station controller, a base station (GBTS/eGBTS/NodeB/co-MPT base station) acts as the initiator to activate the IP PM.

IP PM configuration and performance counters are the same for the eGBTS, NodeB, and co-MPT multimode base station, but those for the GBTS are different.

4.2.2.2.1 GBTS

You can run the GBTS MML command **ACT BTSIPPM** to activate IP PM from the GBTS to a base station controller. Table 4-4 describes the items you need to configure in this command.

Table 4-4 Items and configuration principles for activating IP PM on the GBTS

Item	Description
Detected data stream	For details on the definition of a detected data stream, see 4.1.2.1 Detected Data Stream. ● SIP and DIP of the detected data stream can be set using the **SET BTSIP** command. ● DSCP of the detected data stream is specified by the ***SRVTYPE*** parameter. This parameter specifies the type of data stream on which IP PM detection is to be performed, for example, the CS voice service or CS data service.
Activation direction	Only forward activation can be used and the activation direction is not configurable.
FM packet send period	The sending packet period, or the period for sending FM frames is specified by the ***PMPRD*** parameter.

Item	Description
Reporting an alarm once the predefined packet loss rate threshold is exceeded	The ***LOSTPKTDETECTSW*** parameter specifies whether to enable the function of reporting an alarm once the predefined packet loss rate threshold is exceeded. When this function is enabled and the packet loss rate exceeds the threshold specified by the ***LOSTPKTALARMTHD*** parameter, the GBTS reports ALM-28052 IP Path Excessive Packet Loss Rate.

4.2.2.2.2 eGBTS/NodeB/Co-MPT Multimode Base Station

You can run the MML command **ADD IPPMSESSION** on the eGBTS/NodeB/co-MPT multimode base station to activate IP PM from the eGBTS/NodeB/co-MPT multimode base station to a base station controller. **Table 4-5** describes the items you need to configure in this command.

Table 4-5 Items and configuration principles for activating IP PM on the eGBTS/NodeB/co-MPT multimode base station

Item	Description
Monitored data stream	For details on the definition of a monitored data stream, see **4.1.2.1 Detected Data Stream**. For IP PM on the eGBTS/NodeB/co-MPT multimode base station, you can directly specify a monitored data stream or bind an IP path for that purpose. The ***BINDPATH*** parameter specifies whether to bind an IP path. When the base station is connected to a base station controller, only four-tuples are supported, which include the DSCP. If a data stream is bound to an IP path, you can configure the DSCP in either of the following ways: ● For an IP path of any QoS type, the DSCP of the detected data stream is the DSCP configured for IP PM. ● For an IP path of a specified QoS type, the DSCP of the detected data stream is the DSCP configured for the IP path. You can run the **LST IPPATH** command to check whether an IP path is of a random QoS type or is of a specified QoS type.
Activation direction	The activation direction is specified by the ***DIR*** parameter, whose value can be **UP(Uplink)** or **BIDIR(Bidirection)**. IP PM from the base station to the base station controller only supports uplink activation, and therefore IP PM takes effect only when uplink activation is configured.
FM packet send period	The sending packet period, or the period for sending FM frames, is 100 ms by default. This period is not configurable.

Item	Description
Reporting an alarm once the predefined packet loss rate threshold is exceeded	The function of reporting an alarm once the predefined packet loss rate threshold is exceeded is not configurable, and is enabled by default. When the packet loss rate exceeds the threshold specified by the ***PLRAT*** parameter, ALM-25898 IP Path Excessive Packet Loss Rate is reported. All IP paths for the eGBTS, NodeB, or co-MPT multimode base station use the same packet loss rate threshold for reporting an alarm. You can run the **SET PLRTHRESHOLD** command to specify the threshold.

NOTE

The MML commands and parameters related to IP PM are the same for the eGBTS, NodeB, and co-MPT multimode base station.

4.2.2.3 Networking Scenarios

Table 4-6 describes whether the Abis interface supports IP PM in different networking scenarios.

Table 4-6 Mapping of the Abis interface's capability of supporting IP PM onto networking scenarios

Networking Scenario		Whether IP PM Is Supported	Description
Port	FE/GE/10GE ports are used to transmit data on the BSC and GBTS/eGBTS.	Yes	N/A
Port	E1/T1 ports are used to transmit data on the GBTS/eGBTS.	Yes	From GBSS14.0, IP PM is supported when the GBTS/eGBTS uses IP over E1/T1 transmission and the BSC uses IP over FE/GE transmission. In this case, a GTMUc board must be configured for the GBTS.
Port	E1/T1 ports are used to transmit data on the BSC.	No	IP PM is not supported when the BSC uses IP over E1/T1 transmission.

Networking Scenario		Whether IP PM Is Supported	Description
Link	Ethernet link aggregation groups work in active/standby mode on the BSC side.	Yes	N/A
	Ethernet link aggregation groups work in load sharing mode on the BSC side.	No	N/A
	Ethernet ports work in route-based load sharing mode on the BSC side.	No	IP PM is not supported because route-based load sharing is usually implemented between different boards.
	Ethernet link aggregation groups are used on the eGBTS side.	No	Only the UMPT_G supports Ethernet link aggregation groups.
Board	The UTRPc board works as the transmission interface board for multimode base stations.	Yes	IP PM can be enabled on the UTRPc board in managing mode and on the main control board in non-managing mode.
Site	Base stations are cascaded.	Yes	IP PM can be enabled on each cascaded base station.

Table 4-7 describes whether the Iub interface supports IP PM in different networking scenarios.

Table 4-7 Mapping of the Iub interface's capability of supporting IP PM onto networking scenarios

Networking Scenario		Whether IP PM Is Supported	Description
Port	The RNC and NodeB use IP over FE/GE transmission.	Yes	N/A

Networking Scenario		Whether IP PM Is Supported	Description
	The NodeB uses IP over E1/T1 transmission.	Yes	From RAN14.0, IP PM is supported when the NodeB uses IP over E1/T1 transmission and the RNC uses IP over FE/GE transmission.
	The RNC uses IP over E1/T1 transmission.	No	IP PM is not supported when the RNC uses IP over E1/T1 transmission.
Link	Ethernet link aggregation groups work in active/standby mode on the RNC side.	Yes	N/A
	Ethernet link aggregation groups work in load sharing mode on the RNC side.	No	N/A
	Ethernet ports work in route-based load sharing mode on the RNC side.	No	IP PM is not supported because route-based load sharing is usually implemented between different boards.
Board	The UTRPc board works as the transmission interface board for multimode base stations.	Yes	IP PM can be enabled on the UTRPc board in managing mode and on the main control board in non-managing mode.
Board	The interface board on the RNC works in transmission pool networking mode.	Yes	N/A

Networking Scenario		Whether IP PM Is Supported	Description
	The interface board on the RNC works in transmission pool networking mode, and Ethernet link aggregation groups work in active/standby mode.	Yes	N/A
	The interface board on the RNC works in transmission pool networking mode, and Ethernet link aggregation groups work in intra-board load sharing mode.	Yes	N/A
Site	Base stations are cascaded.	Yes	IP PM can be enabled on each cascaded base station.

NOTE

Inter-board IP PM is not supported, and therefore the ports used to transmit and receive IP PM packets must be on the same board in the preceding networking scenarios.

Bidirectional activation can only be used on the base station controller side of the Iub interface in transmission resource pool networking. It cannot be used on other interfaces. The reasons are as follows:

- Abis interface: The GBTS does not support bidirectional activation. The eGBTS supports bidirectional activation, but the base station controller does not.
- Iub interface: NodeB V2 supports bidirectional activation, but the base station controller does not support bidirectional activation initiated by the base station. Therefore, you can use bidirectional activation only on the base station controller.

4.2.3 IP PM Between Base Stations

This section describes the application of IP PM over the uX2 interface between NodeBs, over the X2/eX2 interface between eNodeBs, over the X2 interface between the gNodeB and eNodeB, and over the Xn/eXn interface between gNodeBs.

IP PM parameter configurations for the NodeB/eNodeB/gNodeB are the same as those for the **eGBTS/NodeB/co-MPT multimode base station**.

4.2.3.1 On the Base Station Side

The NodeB MML command **ADD IPPMSESSION** is used to activate IP PM over the uX2 interface.

The eNodeB MML command **ADD IPPMSESSION** is used to activate IP PM over the X2/eX2 interface.

The gNodeB MML command **ADD IPPMSESSION** is used to activate IP PM over the X2 interface between the gNodeB and eNodeB.

The gNodeB MML command **ADD IPPMSESSION** is used to activate IP PM over the Xn interface.

NOTE

Before activating IP PM on a local base station, ensure that **TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW (5G gNodeB, LTE eNodeB)*** has been set to **ENABLE** on the peer base station. If this parameter is not set to **ENABLE**, the peer base station cannot respond to IP PM sessions.

Table 4-8 Configurations for IP PM on the base station side

Item	Description
Detected data stream	Configurations are the same as those in **Table 4-5** except for the following: The ***IPPMTYPE*** parameter is set to **THREE_TUPLE(THREE_TUPLE)** for IP PM over the uX2 interface because the interface only supports triplets. The ***IPPMTYPE (5G gNodeB, LTE eNodeB)*** parameter is set to **THREE_TUPLE(THREE_TUPLE)** or **FOUR_TUPLE(FOUR_TUPLE)** for IP PM over the X2/eX2/Xn interface because the interface only supports triplets and four-tuples.
Activation direction	The activation direction is specified by the ***DIR (5G gNodeB, LTE eNodeB)*** parameter. Uplink activation is recommended.
FM packet send period	See **Table 4-5**.
Reporting an alarm once the predefined packet loss rate threshold is exceeded	See **Table 4-5**.

Item	Description
Packet loss detection for IPv6 IP PM sessions	This function is controlled by the **TRANSFUNCTIONSW**.***IPPMIPV6PKTLOSDETSW (5G gNodeB, LTE eNodeB)*** parameter, whose default value is **DISABLE**. Changing the parameter setting will cause service exceptions during IPv6 IP PM renegotiation, affecting all inter-base-station coordination and data split services established using IPv6. Services can be initiated again only after IPv6 IP PM negotiation is normal.

NOTE

In endpoint mode, if the uX2 interface, the X2 interface between eNodeBs, the eX2 interface, or the X2 interface between the eNodeB and gNodeB carry coordination services, uplink IP PM sessions are automatically created over the interface. Under these circumstances, bidirectional IP PM sessions cannot be created and activated over the interface. Otherwise, the interface cannot be used.

4.2.3.2 Networking Scenarios

Table 4-9 describes whether the uX2/X2/eX2/Xn interface supports IP PM in different networking scenarios.

Table 4-9 IP PM capability of the uX2/X2/eX2 interface in different networking scenarios

Networking Scenario		Supported or Not	Description
Port	FE/GE/10GE ports are used to transmit data.	Yes	N/A
Board	The UTRPc board works as the transmission interface board.	Yes	N/A
	The UTRPc board works as the transmission interface board for multimode base stations.	Yes	IP PM can be enabled on the UTRPc board in managing mode and on the main control board in non-managing mode.

Networking Scenario		Supported or Not	Description
Link	Ethernet link aggregation groups are used on the ports.	Yes	N/A
Site	Base stations are cascaded.	Yes	IP PM can be enabled on each cascaded base station.

4.2.4 IP PM Between the Base Station and CN

This section describes the application of IP PM over the S1 interface between the eNodeB and CN. The IP PM over the S1 interface can be activated either on the eNodeB or CN side. You need to ensure that the CN devices are provided by Huawei and support IP PM before activating IP PM over the S1 interface.

4.2.4.1 On the eNodeB Side

Before activating IP PM over the S1 interface on the eNodeB side, ensure that the global IP PM switch on the CN side has been turned on.

IP PM over the S1 interface can be manually or automatically configured on the eNodeB side:

- Manual configuration is recommended when some CN devices are not provided by Huawei or when only the DSCPs of one or more links need to be monitored.

  When manually configuring IP PM over the S1 interface, operators are advised to set **IPPMSESSION**.***IPPMTYPE*** to **FOUR_TUPLE(FOUR_TUPLE)**, and set **IPPMSESSION**.***IPPMDSCP*** to the DSCPs of common service types, and operators can only set **IPPMSESSION**.***DIR*** to **UP(Uplink)**.
- Automatic configuration is recommended when all CN devices are provided by Huawei and the eNodeB transmission is configured in endpoint mode. When automatic configuration is applied, IP PM over the S1 interface supports self-management. Specifically, IP PM sessions can be adaptively established and deleted within the specifications provided in **Table 4-10** based on the link establishment and deletion on the user plane. This reduces manual intervention.

  To enable automatic configuration of IP PM over the S1 interface, run the MML command **LST S1** to query the value of **S1**.***UpEpGroupId***. Then, run the MML command **ADD EPGROUP** or **MOD EPGROUP** with endpoint group ID set to the queried value and **EPGROUP**.***IPPMSWITCH*** set to **ENABLE** to enable automatic configuration of IP PM over the user plane of the S1 interface. Under these circumstances, uplink activation is automatically used as the value of the activation direction, and four-tuples is automatically used to specify a detected data stream over the S1 interface on the eNodeB. In addition, operators are advised to set **EPGROUP**.***IPPMDSCP*** to the DSCPs of common service types. Modifying the value of this parameter leads to automatic deletion and reestablishment of IP PM sessions.

In secure networking mode, when **EPGROUP**.***PACKETFILTERSWITCH*** is set to **ENABLE(Enable)**, packet filtering ACL rules for IP PM can be automatically established based on the ACL rule specifications. No manual configuration is required. Automatic configuration of IP PM over the S1 interface can be applied only when all CN devices are provided by Huawei. When some CN devices are not provided by Huawei, the application of automatic configuration results in IP PM activation failures on links for CN devices not provided by Huawei.

NOTE

- After **EPGROUP**.***IPPMSWITCH*** is set to **ENABLE**, the eNodeB periodically establishes IP PM sessions based on the adaptively established connections over the S1-U interface. Operators can wait 1 minute and then run the **LST IPPMSESSION** or **DSP IPPMSESSION** command to query the activation of automatic configuration of IP PM. If **IPPMSESSION**.***CTRLMODE*** is set to **AUTO_MODE** for the IP PM session with **IPPMSESSION**.***IPPMSN*** ranging from 70000 to 79999, IP PM has been activated on the eNodeB side.
- The main control board must be a UMPT or UMDU to support the automatic configuration of IP PM.
- The automatic configuration of IP PM is not supported in simplified endpoint mode as the automatically established EP group cannot be manually modified.

It is recommended that either manual or automatic configuration of IP PM be used over the S1 interface on the eNodeB side. The use of only one configuration mode ensures the effectiveness of the IP PM specifications provided in **Table 4-10** and prevents the repeated establishment of IP PM sessions.

4.2.4.2 On the CN Side

For details on how to activate IP PM on the CN side, see related description in CN product documentation. For details on the requirements on the CN, see **4.4.4 Others**.

NOTE

Before activating IP PM on the CN side, ensure that **TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW (LTE eNodeB, 5G gNodeB)*** has been set to **ENABLE** on the base station. If this parameter is not set to **ENABLE**, the base station cannot respond to IP PM sessions.

4.3 Network Analysis

4.3.1 Benefits

IP PM reduces O&M cost. Specifically, it provides the following benefits:

- Monitoring network performance in real time

  IP PM monitors network performance in real time and provides statistics of transmission counters. This helps operators quickly identify transmission faults and take effective measures, such as expanding network capacity or optimizing the transport network.

- Locating the transmission fault quickly

  IP PM helps quickly locate transmission performance-related problems (such as an overly high packet loss rate and long delay) and isolate the faults, thereby improving network maintainability and reducing O&M cost.
- Reporting an alarm once the predefined packet loss rate threshold is exceeded

  With IP PM, the NE reports an alarm upon detecting that the realtime packet loss rate has exceeded a predefined threshold. This helps operators take rectifying measures in a timely manner.

4.3.2 Impacts

The impacts between this function and other functions are described in **4.4.2.3 Function Impacts**.

With IP PM, FM and BR frames are periodically sent and received between the IP PM initiator and responder. This occupies extra transmission bandwidth and increases the overhead. For example, an extra 6 kbit/s bandwidth is occupied for an IP PM session when the measurement period is 100 ms.

4.4 Requirements

4.4.1 Licenses

None

4.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM	Abis over IP	None	*IPv4 Transmission*	None
UMTS	IP Transmission Introduction on Iub Interface	None	*IPv4 Transmission*	BSC6900 in non-transmission resource pool networking

RAT	Function Name	Function Switch	Reference	Description
	Iub IP Transmission Based on Dynamic Load Balance	None	*Transmission Resource Pool in RNC*	BSC6900/ BSC6910/BSC6960 in transmission resource pool networking

4.4.2.2 Mutually Exclusive Functions

None

4.4.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE	Self-configuration	None	*Automatic OMCH Establishment*	With the Self-configuration or LTE and NR X2 Interface Self-Configuration (LTE FDD) feature, operators can choose not to configure the peer user-plane IP address on the eNodeB side during deployment. However, under these circumstances, IP PM cannot be activated on the eNodeB side. Operators can obtain the peer IP address and then activate IP PM only after the transmission link is automatically established for the eNodeB.
	LTE and NR X2 Interface Self-Configuration (LTE FDD)	**LTE_NR_X2_SON_SETUP_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter	*X2 and S1 Self-Management in NSA Networking*	

RAT	Function Name	Function Switch	Reference	Description
	LTE and NR X2 Interface Self-Configuration (LTE TDD)	**LTE_NR_X2_SON_SETUP_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter	*X2 and S1 Self-Management in NSA Networking*	With the LTE and NR X2 Interface Self-Configuration feature, operators can choose not to configure the peer user-plane IP address on the gNodeB side during deployment. However, under these circumstances, IP PM cannot be activated on the gNodeB side. Operators can obtain the peer user-plane IP address and then activate IP PM only after the transmission link is automatically established for the gNodeB.
	NSA networking based on EPC	**NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*	If the IP PM function is abnormal, the base station cannot obtain the X2 interface delay. As a result, the downlink data split performance of NSA DC is poor or downlink data split fails.

RAT	Function Name	Function Switch	Reference	Description
NR	LTE and NR X2 Interface Self-Configuration (NR)	**X2SON_SETUP_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter	*X2 and S1 Self-Management in NSA Networking*	With the LTE and NR X2 Interface Self-Configuration feature, operators can choose not to configure the peer user-plane IP address on the gNodeB side during deployment. However, under these circumstances, IP PM cannot be activated on the gNodeB side. Operators can obtain the peer user-plane IP address and then activate IP PM only after the transmission link is automatically established for the gNodeB.
	NSA networking based on EPC	**NRCellAlgoSwitch**.***NsaDcSwitch***	*EPC-based NSA Basics*	If the IP PM function is abnormal, the base station cannot obtain the X2 interface delay. As a result, the downlink data split performance of NSA DC is poor or downlink data split fails.

4.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite
- BSC6900/BSC6910/BSC6960

Boards

NE	Board	Supports IP PM or Not
BSC6900	GOUa/FG2a/GOUc/GOUe/FG2c/FG2d/FG2e/GOUd	Yes
BSC6910	EXOUa/EXOUb/GOUc/GOUe/FG2c/FG2d/FG2e/GOUd/GOUf	Yes
BSC6960	CMPU	Yes
GBTS	GTMUc	Yes
eGBTS/NodeB/eNodeB/gNodeB/Co-MPT multimode base stations	UMPT/UMDU/MDUC/UTRPc	Yes

RF Modules

N/A

4.4.4 Others

The requirements on the CN are as follows:

- The CN devices must be provided by Huawei.
- The CN cannot be a cloud core network.

4.5 Operation and Maintenance

4.5.1 When to Use

IP PM can be enabled to serve as a routine measure for monitoring changes in transmission rates and link performance. This facilitates transmission performance assessment and transmission fault diagnosis. Alternatively, enable IP PM when you need to locate data service performance issues, such as unstable download rates. In this case, you can disable IP PM after troubleshooting.

The maximum numbers of IP PM sessions supported by the base stations and base station controllers are listed in **Table 4-10**.

Table 4-10 Maximum number of IP PM sessions supported by NEs

NE	Maximum Number of Supported IP PM Sessions
BSC/RNC	● GOUa/FG2a: Each board supports a maximum of 500 IP PM sessions when serving as the IP PM initiator, and supports a maximum of 1200 IP PM sessions when serving as the IP PM responder. ● EXOUa/EXOUb: 6000 ● GOUc/GOUe/FG2c/FG2d/FG2e/GOUd/GOUf: 4000 ● CMPU: 1200
GBTS	● GTMUc: 8 ● GBTS: 8
eGBTS/NodeB/eNodeB/gNodeB/Co-MPT multimode base stations	● Boards UMPT/UMDU/UTRPc: 128 ● Base stations – Base station where the UMDU serves as the main control board: 128 – Base station where the UMPT serves as the main control board: 500

NOTE

The number of IP PM sessions supported by the base station controller is much greater than that supported by the base station. If the number of IP PM sessions to be activated between a base station controller and a base station exceeds the capability of the base station or its transmission board, excess sessions will not be activated. If the number of IP PM sessions to be activated exceeds the capability of the transmission board of the base station controller, excess sessions will not be activated, and the BSC6900/BSC6910 reports ALM-21396 IP PM Capability Exceeded and the BSC6960 reports ALM-21694 IP PM Capability Exceeded.

Unless otherwise specified, the same IP PM session specifications provided in the preceding table are applied to the board or NE, regardless of whether the board or NE serves as the IP PM initiator or IP PM responder.

When activating IP PM sessions between the base station controller and base station on the base station controller side, operators should also enable the IP PM admission function on the base station side. If the IP PM admission function is not enabled, IP PM sessions fail to be activated on the base station controller side, and the BSC6900/BSC6910 reports ALM-21341 IP PM Activation Failure and the BSC6960 reports ALM-21693 IP PM Activation Failure. To enable this function, for a GBTS, run the **SET BTSGTRANSPARA** MML command on the BSC with **IPPM Admittance** set to **ENABLE**. For other base stations, run the **SET TRANSFUNCTIONSW** MML command on the base station with **IP PM Passive Activation Switch** set to **ENABLE**.

4.5.2 Precautions

- The peer (base station's) mask of an IP path configured on the RNC side must be **255.255.255.255**.
- The eGBTS, NodeB, eNodeB, gNodeB, and co-MPT multimode base station support IP PM detection based on triplets. Before activating IP PM detection based on triplets, ensure that both local and peer ends support IP PM detection based on triplets.
- The BSC6960 supports IP PM detection based on triplets. Before activating IP PM detection based on triplets, ensure that the peer end supports IP PM detection based on triplets.
- The intermediate transmission equipment cannot change the DSCP value carried in detected packets during the IP PM detection based on four-tuples.
- Bidirectional activation of IP PM is not recommended for the eNodeB and gNodeB. The reasons are as follows:
  - For the S1 interface, the current version of the S-GW does not allow the eNodeB to initiate bidirectional activation. Therefore, the S1 interface cannot be configured with bidirectional IP PM activation.
  - For the X2 interface, when both ends are configured with IP PM, an activation conflict occurs and a certain IP PM session fails to be activated if either end is configured with bidirectional IP PM activation.
  - For the eX2 interface, if either end is configured with bidirectional IP PM activation, a conflict with the session automatically set up by the eNodeB may occur. As a result, the eX2 interface cannot work normally.
  - For the eXn interface, gNodeBs will automatically create IP PM sessions to detect link status. If an IP PM session of bidirectional activation type is configured at either end of the eXn interface, it may conflict with an IP PM session automatically created by the gNodeB. As a result, the eXn interface cannot work properly.
  - For the eX2/eXn interface, when an IP PM session that supports delay measurement with microsecond precision needs to be configured, four-tuples must be used. If triplets are used, the IP PM session does not support delay measurement with microsecond precision.
- The NodeB and eNodeB support only forward activation in IP PM V14.
- Before activating IP PM on the base station, ensure that **TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW (5G gNodeB, LTE eNodeB)*** has been set to **ENABLE**. If this parameter is not set to **ENABLE**, the base station cannot respond to IP PM sessions.

4.5.3 Data Configuration

4.5.3.1 Data Preparation

Table 4-11 and **Table 4-12** list the data to prepare before activating IP PM on the BSC6900 in non-transmission resource pool networking and on the BSC6900/BSC6910/BSC6960 in transmission resource pool networking, respectively. **Table 4-13** lists the data to prepare before activating IP PM on the GBTS, and **Table 4-14** lists the data to prepare before activating IP PM on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station.

Table 4-11 Data to prepare before activating IP PM on the BSC6900 in non-transmission resource pool networking

Parameter Name	Parameter ID	Setting Notes
Adjacent Node ID	**IPPM**.***ANI***	Set this parameter to the ID of the peer end.
IP path ID	**IPPM**.***PATHID***	Set this parameter to the ID of the IP path to be detected.
IS QOSPATH	**IPPM**.***ISQOSPATH***	Set this parameter to a value consistent with the type of the IP path.
PHB	**IPPM**.***PHB***	Set this parameter based on the service that needs to be detected using IP PM only when the **IPPM**.***ISQOSPATH*** parameter is set to **YES(YES)**. **NOTE** Run the **LST ADJMAP** command to query the TRMMAP index used by the base station. If no result is found, the default Abis interface TRMMAP is used, and the TRMMAP index is 10. Then, run the **LST TRMMAP** command to query the PHB used by the corresponding base station service.
IPPM packet send period	**IPPM**.***PMPRD***	You are advised to set this parameter to the default value.
Excessive Packet Loss Alarm Switch	**IPPM**.***LOSTPKTDETECTSW***	You are advised to set this parameter to **ON**.
Lost-Packet Alarm Threshold	**IPPM**.***LOSTPKTALARMTHD***	You are advised to set this parameter to the default value.

Table 4-12 Data to prepare before activating IP PM on the BSC6900/BSC6910/BSC6960 in transmission resource pool networking

Parameter Name	Parameter ID	Setting Notes
Adjacent Node ID	**IPPOOLPM**.***ANI***	N/A
Per-Hop Behavior	**IPPOOLPM**.***PHB***	Set this parameter based on the **TRMMAP** MO.

Parameter Name	Parameter ID	Setting Notes
FM packet send period[100ms]	**IPPOOLPM**.***PMPRD***	You are advised to set this parameter to the default value.
Excessive Packet Loss Alarm Switch	**IPPOOLPM**.***LOSTPKTDETECTSW***	You are advised to set this parameter to **ON**.
Lost-Packet Alarm Threshold	**IPPOOLPM**.***LOSTPKTALARMTHD***	You are advised to set this parameter to the default value.
Sip Type	**IPPOOLPM**.***SIPTYPE***	N/A
IP Address in Transmission Resource Pool	***IPADDR***	Set this parameter only when the ***SIPTYPE*** parameter is set to **ASSIGN_ONE_SIP**.
Activation Direction	**IPPOOLPM**.***DR***	For GSM, set this parameter to **SOURCE** because GSM supports only downlink transmission QoS detection. For UMTS, the recommended values are as follows: ● Set this parameter to **SINK** to detect the uplink transmission QoS. ● Set this parameter to **SOURCE** to detect the downlink transmission QoS. ● Set this parameter to **BOTH** to detect the uplink and downlink transmission QoS.

Table 4-13 Data to prepare before activating IP PM on the GBTS

Parameter Name	Parameter ID	Setting Notes
IPPM Admittance	**BTSGTRANSPARA**.***IPPMADMITTANCE***	Whether to enable the IP PM admission function. When this parameter is set to **ENABLE**, only the user-plane IP address of the base station responds to the IP PM activation request sent by the peer end. When this parameter is set to **DISABLE**, all IP addresses do not respond to the IP PM activation request sent by the peer end. The peer end cannot activate IP PM sessions.
Service Type	**BTSIPPM**.***SRVTYPE***	N/A
BTS Index	**BTSIPPM**.***BTSID***	N/A
IPPM Packet Send Period	**BTSIPPM**.***PMPRD***	Set this parameter to the default value.
Excessive Packet Loss Alarm Switch	**BTSIPPM**.***LOSTPKTDETECTSW***	You are advised to set this parameter to **ON**.
Lost-Packet Alarm Threshold	**BTSIPPM**.***LOSTPKTALARMTHD***	Set this parameter to the default value.

Table 4-14 Data to prepare before activating IP PM on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station

Parameter Name	Parameter ID	Setting Notes
IP PM Passive Activation Switch	**TRANSFUNCTIONSW**.***IPPMPASSIVEACTIVATIONSW (5G gNodeB, LTE eNodeB)***	Whether to enable IP PM passive activation. When this parameter is set to **ENABLE**, only the user-plane IP address responds to the IP PM activation request sent by the peer end. The peer end can activate IP PM sessions. When this parameter is set to **DISABLE**, all IP addresses do not respond to the IP PM activation request sent by the peer end. The peer end cannot activate IP PM sessions.
Session ID	**IPPMSESSION**.***IPPMSN (5G gNodeB, LTE eNodeB)***	N/A

Parameter Name	Parameter ID	Setting Notes
IP Version	**IPPMSESSION**.***IPVERSION (LTE eNodeB, 5G gNodeB)***	Indicates the IP protocol version used by IP PM.
IP PM Type	**IPPMSESSION**.***IPPMTYPE (5G gNodeB, LTE eNodeB)***	Generally, **FOUR_TUPLE(FOUR_TUPLE)** is recommended for this parameter. However, if you want to deploy IP PM over the X2/eX2/Xn/eXn interface and the intermediate transmission equipment changes the DSCP value carried in packets, set this parameter to **THREE_TUPLE(THREE_TUPLE)**. If the ***IPVERSION (LTE eNodeB, 5G gNodeB)*** parameter is set to **IPv6**, this parameter must be set to **THREE_TUPLE(THREE_TUPLE)**.
Bind IP Path	**IPPMSESSION**.***BINDPATH (5G gNodeB, LTE eNodeB)***	If IP PM measures service flows carried on a bound IP path, set this parameter to **YES**. ● If the ***IPPMTYPE (5G gNodeB, LTE eNodeB)*** parameter is set to **FOUR_TUPLE(FOUR_TUPLE)**, IP PM sessions can be bound to IP paths whose type is fixed or any. ● If the ***IPPMTYPE (5G gNodeB, LTE eNodeB)*** parameter is set to **THREE_TUPLE(THREE_TUPLE)**, IP PM sessions can be bound only to IP paths whose type is any. If IP PM measures service flows carried on a certain IP address, set this parameter to **NO**. If the ***IPVERSION (LTE eNodeB, 5G gNodeB)*** parameter is set to **IPv6**, no IP path can be bound.
IP Path ID	**IPPMSESSION**.***PATHID (5G gNodeB, LTE eNodeB)***	Set this parameter only when the ***BINDPATH (5G gNodeB, LTE eNodeB)*** parameter is set to **YES**.
Local IP	**IPPMSESSION**.***LOCALIP (5G gNodeB, LTE eNodeB)***	Set this parameter to the IP address of the base station.
Peer IP	**IPPMSESSION**.***PEERIP (5G gNodeB, LTE eNodeB)***	Set this parameter to the IP address of the peer side.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	**IPPMSESSION**.***LOCALIPV6 (LTE eNodeB, 5G gNodeB)***	Indicates the local IPv6 address of an IP PM session. This parameter takes effect only for LTE and NR.
Peer IPv6 Address	**IPPMSESSION**.***PEERIPV6 (LTE eNodeB, 5G gNodeB)***	Indicates the peer IPv6 address of an IP PM session. This parameter takes effect only for LTE and NR.
DSCP	**IPPMSESSION**.***IPPMDSCP (5G gNodeB, LTE eNodeB)***	Set this parameter to the DSCP value for the data stream on which IP PM is to be performed. It is recommended that this parameter be set to the DSCP values of common service types.
Activate Direction	**IPPMSESSION**.***DIR (5G gNodeB, LTE eNodeB)***	Set this parameter to **UP(Uplink)**. If the ***IPVERSION (LTE eNodeB, 5G gNodeB)*** parameter is set to **IPv6**, this parameter must be set to **UP(Uplink)**.
IP PM IPv6 Pkt Loss Detect Switch	**TRANSFUNCTIONSW**.***IPPMIPV6PKTLOSDETSW (5G gNodeB, LTE eNodeB)***	You are advised to set this parameter to **DISABLE**.

Table 4-15 Data to prepare before activating the automatic configuration of IP PM over the S1 interface on the eNodeB/co-MPT multimode base station

Parameter Name	Parameter ID	Setting Notes
IP PM Automatic Setup and Deletion Switch	**EPGROUP**.***IPPMSWITCH (5G gNodeB, LTE eNodeB)***	N/A
IP PM DSCP	**EPGROUP**.***IPPMDSCP (5G gNodeB, LTE eNodeB)***	N/A

4.5.3.2 Using MML Commands

Before using MML commands, refer to **4.4 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Activating forward IP PM in non-transmission resource pool networking on the BSC6900

```
ACT IPPM: ANI=0, PATHID=1, ISQOSPATH=NO, PMPRD=1, LOSTPKTDETECTSW=OFF;
```

Activating forward IP PM in transmission resource pool networking on the BSC6900/BSC6910 (the Abis interface used as an example)

```
ACT IPPOOLPM: ANI=0, ITFT=ABIS, SIPTYPE=ADJNODE_BIND_SIP, PHB=AF11, DR=SOURCE, PMPRD=1,
LOSTPKTDETECTSW=OFF;
```

Activating backward IP PM in transmission resource pool networking on the BSC6900/BSC6910 (the Abis interface used as an example)

```
ACT IPPOOLPM: ANI=0, ITFT=ABIS, SIPTYPE=ADJNODE_BIND_SIP, PHB=AF11, DR=SINK, PMPRD=1,
LOSTPKTDETECTSW=OFF;
```

Activating bidirectional IP PM in transmission resource pool networking on the BSC6900/BSC6910 (the Abis interface used as an example)

```
ACT IPPOOLPM: ANI=0, ITFT=ABIS, SIPTYPE=ADJNODE_BIND_SIP, PHB=AF11, DR=BOTH, PMPRD=1,
LOSTPKTDETECTSW=OFF;
SET TRANSFUNCTIONSW: IPPMPASSIVEACTIVATIONSW=ENABLE;
```

Activating backward IP PM in transmission resource pool networking on the BSC6960 (the Abis interface used as an example)

```
ACT IPPOOLPM: ANI=1, PHB=EF, LOSTPKTDETECTSW=OFF;
```

Activating uplink IP PM on the GBTS

```
ACT BTSIPPM: IDTYPE=BYID, BTSID=9, SRVTYPE=CSVOICE;
```

Activating uplink IP PM on the eGBTS/NodeB/eNodeB/gNodeB

```
ADD IPPMSESSION: IPPMSN=0, IPVERSION=IPv4, IPPMTYPE=FOUR_TUPLE, BINDPATH=NO,
LOCALIP="192.168.11.110", PEERIP="192.168.22.220", IPPMDSCP=20;
ADD IPPMSESSION: IPPMSN=0, IPVERSION=IPv6, IPPMTYPE=THREE_TUPLE, BINDPATH=NO,
LOCALIPV6="1140:201:10::57", PEERIPV6="1140:201:10::54";
SET TRANSFUNCTIONSW: IPPMIPV6PKTLOSDETSW=DISABLE;
```

Activating the automatic configuration of IP PM over the S1 interface on the eNodeB/co-MPT multimode base station after querying the value of **S1**.***UpEpGroupId*** by running the **LST S1** command (the value **456** used as an example)

```
MOD EPGROUP: EPGROUPID=456,IPPMSWITCH=ENABLE,IPPMDSCP=20;
```

Enabling IP PM admission on the GBTS side

```
SET BTSGTRANSPARA: IPPMADMITTANCE=ENABLE;
```

Turning on the IP PM passive activation switch on the eGBTS/NodeB/eNodeB/gNodeB

```
SET TRANSFUNCTIONSW: IPPMPASSIVEACTIVATIONSW=ENABLE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivating IP PM in non-transmission resource pool networking on the BSC6900

```
DEA IPPM: ANI=0, PATHID=1;
```

Deactivating IP PM in transmission resource pool networking on the BSC6900/ BSC6910

```
DEA IPPOOLPM: ANI=0, ITFT=ABIS, SIPTYPE=ADJNODE_BIND_SIP, PHB=AF12;
```

Deactivating IP PM in transmission resource pool networking on the BSC6960

```
DEA IPPOOLPM: ANI=1, PHB=EF;
```

Deactivating IP PM on the GBTS

```
DEA BTSIPPM: IDTYPE=BYID, BTSID=9, SRVTYPE=CSVOICE;
```

Deactivating IP PM on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station

```
RMV IPPMSESSION: IPPMSN=0;
```

Deactivating the automatic configuration of IP PM over the S1 interface on the eNodeB/co-MPT multimode base station after querying the value of **S1**.***UpEpGroupId*** by running the **LST S1** command (the value **456** used as an example)

```
MOD EPGROUP: EPGROUPID=456,IPPMSWITCH=DISABLE;
```

4.5.4 Activation Verification

You can run an MML command to verify whether IP PM has been successfully activated.

To verify whether IP PM has been successfully activated in non-transmission resource pool networking on the BSC6900, perform the following step:

Step 1 Run the **DSP IPPM** command to query the measurement parameters related to IP PM. IP PM has been successfully activated if the value of **IPPM State** is **PM_UP**.

----End

To verify whether IP PM has been successfully activated in transmission resource pool networking on the BSC6900/BSC6910/BSC6960, perform the following step:

Step 1 Run the **DSP IPPOOLPM** command to query the measurement parameters related to IP PM.

IP PM has been successfully activated if the value of **Detection State** is **PM_UP** in the information about the following items:

- **Source PM State** (forward IP PM activation)
- **Sink PM State** (backward IP PM activation)
- **Source PM State** and **Sink PM State** (bidirectional IP PM activation)

----End

To verify whether IP PM has been successfully activated on the GBTS, perform the following step:

Step 1 Run the BSC MML command **DSP BTSIPPMLNK** to query the measurement parameters related to IP PM. IP PM has been successfully activated if the value of **PM Active State** is **Activated** in the command output.

----End

To verify whether IP PM has been successfully activated on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station, perform the following step:

Step 1 Run the **DSP IPPMSESSION** command to query the measurement parameters. IP PM has been successfully activated if the value of **Activate State(Up)** is **IP PM UP** in the command output.

----End

NOTE

When the Abis interface works in the IP pool networking, certain services are needed before you can observe the IP PM state.

Then, you can run the **DSP IPPMSESSION** command on the base station side or **DSP IPPM** command on the base station controller side to check the state of IP PM, which can be either of the following:

- IDLE: IP PM is not activated.
- ACTING: Negotiation for IP PM is going on.
- PMUP: IP PM is successfully activated.
- DEACTING: IP PM is being deactivated.

4.5.5 Network Monitoring

4.5.5.1 Routine Monitoring

IP PM is recommended for monitoring online service changes in real time. This facilitates collecting traffic statistics and diagnosing transmission faults.

IP PM-related performance counters can be used to assess IP network performance. These performance counters can be monitored based on actual network requirements. Generally, if no specific requirements are imposed, first monitor the delay and packet loss rate, and then the delay variation. Note that the average values of these items should be paid attention to and the maximum values should be used as a reference.

NOTE

For IP PM, the delay and delay variation are measured based on test packets, whereas the packet loss rate is measured based on services. There is a service flow on the X2 interface only when a handover is being performed. Therefore, the delay and delay variation of the X2 interface can be measured, but the packet loss rate may be 0 or an extremely small value during a measurement period.

For details about QoS requirements in actual application scenarios, see *Transmission Resource Management Feature Parameter Description*.

IP PM performance measurement on the base station controller side is classified as follows:

- IP PM in non-transmission resource pool networking (IP PM performance measurement based on IP paths): indicates IP PM performance when IP paths are used.
- IP PM in transmission resource pool networking (IP PM performance measurement based on the transmission resource pool): indicates IP PM performance in the transmission resource pool.

NOTE

If IP PM is successfully activated, the value of **VS.IPPOOL.IPPM.Rtt.Means** is not 0. In RNC in pool scenarios, the values of counters such as **VS.IPPOOL.IPPM.Peer.RxPkts** may be 0 for the backup RNC, which is normal.

IP PM performance measurement on the base station side is classified as follows:

- IP PM related performance counters on the GBTS
- IP PM related performance counters on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station

Table 4-16, **Table 4-17**, and **Table 4-18** list the IP PM related performance counters of the base station and base station controller.

Table 4-16 IP PM related performance counters (eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station)

Measurement Item	IP PM Related Performance Counter
Round trip delay	**VS.IPPM.Rtt.Means (LTE eNodeB, 5G gNodeB)** **VS.IPPM.MaxRttDelay (LTE eNodeB, 5G gNodeB)** **VS.IPPM.MinRttDelay (LTE eNodeB, 5G gNodeB)** **VS.IPPM.HighLevel.Rtt.Means (LTE eNodeB, 5G gNodeB)** **VS.IPPM.HighLevel.MaxRttDelay (LTE eNodeB, 5G gNodeB)** **VS.IPPM.HighLevel.MinRttDelay (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Owd.Means (LTE eNodeB, 5G gNodeB)** **VS.IPPM.MaxOwdDelay (LTE eNodeB, 5G gNodeB)** **VS.IPPM.MinOwdDelay (LTE eNodeB, 5G gNodeB)**

Measurement Item	IP PM Related Performance Counter
Transmission rate	**VS.IPPM.Bits.MeansTx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peak.Bits.RateTx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peer.Bits.MeansRx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peer.Peak.Bits.RateRx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Pkts.MeansTx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peak.Pkts.RateTx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peer.Pkts.MeansRx (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peer.Peak.Pkts.RateRx (LTE eNodeB, 5G gNodeB)**
Packet loss rate	**VS.IPPM.Forword.DropMeans (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Forword.Peak.DropRates (LTE eNodeB, 5G gNodeB)**
Delay variation	**VS.IPPM.Forward.JitterStandardDeviation (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Back.JitterStandardDeviation (LTE eNodeB, 5G gNodeB)**
Number of packets	**VS.IPPM.Local.TxPkts (LTE eNodeB, 5G gNodeB)** **VS.IPPM.Peer.RxPkts (LTE eNodeB, 5G gNodeB)**

Table 4-17 IP PM related performance counters (GBTS)

Measurement Item	IP PM Related Performance Counter
Round trip delay	**BTS.IPPM.Rtt.Means** **BTS.IPPM.MaxRttDelay**
Transmission rate	**BTS.IPPM.Bits.MeansTx** **BTS.IPPM.Peak.Bits.RateTx** **BTS.IPPM.Peer.Bits.MeansRx** **BTS.IPPM.Peer.Peak.Bits.RateRx** **BTS.IPPM.Peer.Peak.Pkts.RateRx** **BTS.IPPM.Peak.Pkts.RateTx** **BTS.IPPM.Peer.Pkts.MeansRx** **BTS.IPPM.Peer.Peak.Pkts.RateRx**
Packet loss rate	**BTS.IPPM.Forword.DropMeans** **BTS.IPPM.Forword.Peak.DropRates**

Measurement Item	IP PM Related Performance Counter
Delay variation	**BTS.IPPM.Forward.JitterStandardDeviation** **BTS.IPPM.Back.JitterStandardDeviation**
Number of packets	N/A

Table 4-18 IP PM related performance counters (BSC6900/BSC6910/BSC6960)

Measurement Item	IP PM Related Performance Counter (Non-Transmission Resource Pool Networking)	IP PM Related Performance Counter (Transmission Resource Pool Networking)
Round trip delay	**VS.IPPM.Rtt.Means** **VS.IPPM.MaxRttDelay**	**VS.IPPOOL.IPPM.Rtt.Means** **VS.IPPOOL.IPPM.MaxRttDelay**
Transmission rate	**VS.IPPM.Bits.MeansTx** **VS.IPPM.Peak.Bits.RateTx** **VS.IPPM.Peer.Bits.MeansRx** **VS.IPPM.Peer.Peak.Bits.RateRx** **VS.IPPM.Pkts.MeansTx** **VS.IPPM.Peak.Pkts.RateTx** **VS.IPPM.Peer.Pkts.MeansRx** **VS.IPPM.Peer.Peak.Pkts.RateRx**	**VS.IPPOOL.IPPM.ForwardBits.MeansTx** **VS.IPPOOL.IPPM.Peak.ForwardBits.RateTx** **VS.IPPOOL.IPPM.ForwardPkts.MeansTx** **VS.IPPOOL.IPPM.Peak.ForwardPkts.RateTx** **VS.IPPOOL.IPPM.BackwardBits.MeansTx** **VS.IPPOOL.IPPM.Peak.BackwardBits.RateTx** **VS.IPPOOL.IPPM.BackwardPkts.MeansTx** **VS.IPPOOL.IPPM.Peak.BackwardPkts.RateTx**
Packet loss rate	**VS.IPPM.Forword.DropMeans** **VS.IPPM.Forword.Peak.DropRates** **VS.IPPM.Forward.Precise.Peak.DropRates**	**VS.IPPOOL.IPPM.Forward.DropMeans** **VS.IPPOOL.IPPM.Forward.Peak.DropRates** **VS.IPPOOL.IPPM.Backward.DropMeans** **VS.IPPOOL.IPPM.Backward.Peak.DropRates**

Measurement Item	IP PM Related Performance Counter (Non-Transmission Resource Pool Networking)	IP PM Related Performance Counter (Transmission Resource Pool Networking)
Delay variation	**VS.IPPM.Forward.JitterStandardDeviation** **VS.IPPM.Back.JitterStandardDeviation**	**VS.IPPOOL.IPPM.Forward.JitterMax** **VS.IPPOOL.IPPM.Forward.JitterStandardDeviation**
Number of packets	N/A	N/A

4.5.5.2 Locating Faults

Operators can use IP PM for locating a fault through either of the following ways:

- Excessive packet loss alarm

  If an excessive packet loss alarm is reported during IP PM, there is a fault on transmission links, which may result in a high packet loss rate.
- Performance counter monitoring

  Values of IP PM performance counters are compared with QoS requirements during monitoring. If the values are less than what are required by QoS, the transmission link is functioning normally. If the values are greater than what are required, a fault may occur on the transmission link.

  NOTE

  When an eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station supports IP PM V14 and V15 in negotiation, the average delay in the real-time performance monitoring results on the LMT and MAE is accurate to microseconds. Under other circumstances, the average delay is accurate to milliseconds. The negotiated IP PM version is indicated by the **IP PM Protocol Version** parameter in the **DSP IPPMSESSION** command output.

Performance counters related to IP PM are listed in **Table 4-16**.

For the BSC6900/BSC6910, check the real-time performance monitoring results on the LMT for IP PM on the BSC/RNC as follows:

Step 1 Log in to the LMT and click **Monitor**. The **Monitor** tab page is displayed.

Step 2 In the **Monitor** navigation tree, choose **Monitor** > **Common Monitoring** > **Link Performance Monitoring**.

Step 3 In the displayed **Link Performance Monitoring** dialog box, start a monitoring task.

- For non-transmission resource pool networking, Choose **IPPM** from the **Monitor Item** drop-down list box and set other parameters to appropriate values. Then, click **Submit**.
- Choose **IPPOOL PM** from the **Monitor Item** drop-down list box and set other parameters to appropriate values. Then, click **Submit**.

Step 4 Check real-time performance monitoring results.

----End

For the BSC6960, the real-time performance monitoring results cannot be checked on the LMT for IP PM on the BSC/RNC.

Check the real-time performance monitoring results on the LMT for IP PM on the eGBTS/NodeB/eNodeB/co-MPT multimode base station (link mode) as follows:

Step 1 Log in to the LMT and click **Monitor**. The **Monitor** tab page is displayed.

Step 2 In the **Monitor** navigation tree, choose **Monitor** > **Common Monitoring** > **Transport Link Traffic Monitoring**.

Step 3 In the displayed dialog box, click **Include IPPM Statistic**, select the IP paths to be detected, and enter the values to **IP Path ID**. Then, click **Submit**.

Step 4 Double-click the corresponding task to check the IP PM measurement results.

----End

Check real-time performance monitoring results on the LMT for IP PM on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station (endpoint mode) as follows:

Step 1 Log in to the LMT and click **Monitor**. The **Monitor** tab page is displayed.

Step 2 In the **Monitor** navigation tree, choose **Monitor** > **Common Monitoring** > **Transport Auto Setup User Plane Monitoring**.

Step 3 In the displayed dialog box, set **IPPM flag** to **User Plane Link+IPPM**. The **IPPMSESSION**.***IPVERSION*** parameter is configurable.

- If **IPPMSESSION**.***IPVERSION*** is set to **IPv4**:
  - When **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **THREE_TUPLE**, enter the virtual route instance, source IP address, and destination IP address to be checked, and then click **Yes**. The source and destination IP addresses must be the same as **IPPMSESSION**.***LOCALIP (LTE eNodeB, 5G gNodeB)*** and **IPPMSESSION**.***PEERIP (LTE eNodeB, 5G gNodeB)***.
  - When **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **FOUR_TUPLE**, enter the virtual route instance, source IP address, destination IP address, and DSCP to be checked, and then click **Yes**. The source IP address, destination IP address, and DSCP must be the same as **IPPMSESSION**.***LOCALIP (LTE eNodeB, 5G gNodeB)***, **IPPMSESSION**.***PEERIP (LTE eNodeB, 5G gNodeB)***, and **IPPMSESSION**.***IPPMDSCP (LTE eNodeB, 5G gNodeB)***.
- If **IPPMSESSION**.***IPVERSION*** is set to **IPv6**, **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **THREE_TUPLE** by default. Enter the virtual route instance, source IPv6 address, and destination IPv6 address to be checked, and then click **Yes**. The source and destination IPv6 addresses must be the same as **IPPMSESSION**.***LOCALIPV6 (LTE eNodeB, 5G gNodeB)*** and **IPPMSESSION**.***PEERIPV6 (LTE eNodeB, 5G gNodeB)***, respectively.

Step 4 Double-click the corresponding task to check the IP PM measurement results.

----End

Check real-time performance monitoring results on the MAE for IP PM on the eGBTS/NodeB/eNodeB/co-MPT multimode base station (link mode) as follows:

Step 1 On the MAE, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the left navigation tree on the displayed tab page, choose **Base Station Device and Transport > Transport Performance Monitoring** > **Transport Link Traffic Monitoring**.

Step 3 In the displayed dialog box, set **Extend Option** to **Include IPPM Statistic**, select the IP paths to detect, and enter the values for **PATH ID**. Then, click **Finish**.

Step 4 Double-click the corresponding task to check the IP PM measurement results.

----End

Check real-time performance monitoring results on the MAE for IP PM on the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station (endpoint mode) as follows:

Step 1 On the MAE, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the left navigation tree on the displayed tab page, choose **Base Station Device and Transport > Transport Performance Monitoring** > **Transport Auto Setup User Plane Monitoring**.

Step 3 In the displayed dialog box, set **IPPM flag** to **Include**. The **IPPMSESSION**.***IPVERSION*** parameter is configurable.

- If **IPPMSESSION**.***IPVERSION*** is set to **IPv4**:
  - When **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **THREE_TUPLE**, enter the virtual route instance, source IP address, and destination IP address to be checked, and then click **Finish**. The source and destination IP addresses must be the same as **IPPMSESSION**.***LOCALIP (LTE eNodeB, 5G gNodeB)*** and **IPPMSESSION**.***PEERIP (LTE eNodeB, 5G gNodeB)***.
  - When **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **FOUR_TUPLE**, enter the virtual route instance, source IP address, destination IP address, and DSCP to be checked, and then click **Finish**. The source IP address, destination IP address, and DSCP must be the same as **IPPMSESSION**.***LOCALIP (LTE eNodeB, 5G gNodeB)***, **IPPMSESSION**.***PEERIP (LTE eNodeB, 5G gNodeB)***, and **IPPMSESSION**.***IPPMDSCP (LTE eNodeB, 5G gNodeB)***.
- If **IPPMSESSION**.***IPVERSION*** is set to **IPv6**, **IPPMSESSION**.***IPPMTYPE (LTE eNodeB, 5G gNodeB)*** for the base station is set to **THREE_TUPLE** by default. Enter the virtual route instance, source IPv6 address, and destination IPv6 address to be checked, and then click **Yes**. The source and destination IPv6 addresses must be the same as **IPPMSESSION**.***LOCALIPV6 (LTE eNodeB, 5G gNodeB)*** and **IPPMSESSION**.***PEERIPV6 (LTE eNodeB, 5G gNodeB)***, respectively.

Step 4 Double-click the corresponding task to check the IP PM measurement results.

----End

4.5.5.3 Possible Issues

If IP PM fails to be activated, check whether a firewall shields the UDP port used by IP PM. For details about this port, see the communication matrix documents in the product documentation.

If IP PM fails to be activated or the packet loss rate exceeds a specified threshold during monitoring, the NE reports a corresponding alarm. **Table 4-19** lists these alarms, which vary by NE type.

Table 4-19 Alarm reported if IP PM fails to be activated

NE	Alarm Name
BSC/RNC	BSC6900/BSC6910: ALM-21341 IP PM Activation Failure BSC6960: ALM-21693 IP PM Activation Failure
	BSC6900/BSC6910: ALM-21396 IP PM Capability Exceeded BSC6960: ALM-21694 IP PM Capability Exceeded
	ALM-21352 IP Path Excessive Packet Loss Rate
	BSC6900/BSC6910: ALM-21395 Adjacent Node IP Path Excessive Packet Loss Rate BSC6960: ALM-21695 Adjacent Node IP Path Excessive Packet Loss Rate
GBTS	ALM-28052 IP Path Excessive Packet Loss Rate
eGBTS/NodeB/eNodeB/gNodeB/Co-MPT multimode base station	ALM-25900 IP PM Activation Failure

When automatic configuration of IP PM over the S1 interface is enabled:

- If IP PM using triplets is manually configured for a transmission link of the eNodeB, the IP PM activation fails as the CN does not support IP PM using triplets. The activation of IP PM using four-tuples also fails on this transmission link. Under these circumstances, operators must manually deactivate the manually configured IP PM using triplets.
- If **Packet Filter ACL Rule Auto-Setup-Deletion SW** in the **EPGROUP** MO is turned on, ALM-26245 Configuration Data Inconsistency is reported to alert operators to every establishment failure of packet filtering ACL rules for IP PM due to limited ACL rule specifications.

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *SRAN Networking and Evolution Overview*
- *IP Active Performance Measurement*
- *Automatic OMCH Establishment*
- *IPv4 Transmission*
- *EPC-based NSA Basics*
- *X2 and S1 Self-Management in NSA Networking*
- *Transmission Resource Pool in BSC* in *GBSS Feature Documentation*
- *Transmission Resource Pool in RNC* in *RAN Feature Documentation*
- *Transmission Resource Management* in *RAN Feature Documentation*, *eRAN Feature Documentation*, or *5G RAN Feature Documentation*
- *3900 & 5900 Series Base Station Communication Matrix* in *3900 & 5900 Series Base Station Product Documentation*